

Complete Quant Finance Learning Path

Professional Guide with Resource Links
From Foundation to Interview Ready in 40 Weeks

Amit Kumar Jha

A comprehensive, structured roadmap covering 19 professional resources across Mathematics, Asset Class Models, Derivatives Products, Projects, and Interview Preparation. All materials are desk-oriented, interview-tested, and include lifetime access with updates.

Quick Start Guide - Read This First

How to Use This Document

This learning path is designed for:

- Aspiring quants from mathematics, physics, engineering, computer science, or finance backgrounds
- Career switchers looking to transition into quantitative finance
- Students preparing for quant interviews at tier-1 firms
- Professionals seeking to deepen their understanding of derivatives pricing and risk management

What you'll achieve:

- Master mathematical foundations: Linear Algebra, Differential Equations, Stochastic Calculus, Statistics, ML
- Understand pricing models for all major asset classes: IR, FX, Equity, Credit, Inflation
- Learn 50+ derivative products from vanillas to exotics including hybrids
- Build 43 industry-grade projects demonstrating practical implementation skills
- Prepare for interviews with 250+ solved problems from real quant interviews

How to Navigate This Guide

1. **Start with Phase 1** - Everyone begins here regardless of background
2. **Follow the phase sequence** - Each phase builds on previous knowledge
3. **Use coupon codes** - Apply the codes when purchasing resources for discounts
4. **Track your progress** - Use the success metrics at the end of each phase
5. **Adjust based on background** - See Section 8 for customized paths

Time Commitment & Timeline

Standard Track: 40 weeks (9-10 months) at 20-25 hours/week

Accelerated Track: 24-26 weeks (6 months) at 30-35 hours/week, focusing on 2-3 asset classes

Part-Time Track: 52-60 weeks (12-15 months) at 10-15 hours/week

Intensive Track: 16-20 weeks (4-5 months) for candidates with strong math background

Investment Overview

All resources offer coupon codes (10% discount on each). Purchasing the complete bundle is more cost-effective than individual purchases. Each resource includes:

- Lifetime access with no expiration
- All future updates and additions
- Downloadable PDF format
- Desk-oriented, practical content based on real trading floor experience

4.3 Week-by-Week Breakdown

Weeks 14-15: Interest Rates Models

- Master Hull-White 1-factor: calibration to caps/floors, implementation, mean reversion
- Learn Hull-White 2-factor for decorrelated term structure moves
- Study LIBOR Market Model (LMM) for exotic rates derivatives
- Understand SABR model for capturing swaption volatility smile
- Learn when each model is appropriate and limitations
- Implement basic versions in Python/C++

Weeks 16-17: FX Models

- Master Garman-Kohlhagen model for vanilla FX options
- Learn FX smile modeling: risk reversals and butterflies
- Study stochastic volatility models for FX
- Understand quanto adjustments and cross-currency dynamics
- Learn jump-diffusion models for FX crises
- Study FX carry strategies and their risks

Weeks 18-19: Equity Models

- Understand Black-Scholes assumptions and where they break
- Master local volatility models (Dupire equation)
- Learn Heston model for stochastic volatility
- Study jump-diffusion (Merton, Kou) for crash modeling
- Understand equity skew dynamics and term structure
- Learn dividend modeling and forward pricing

Week 20: Credit Models

- Master intensity-based (reduced-form) models
- Learn structural models (Merton, Black-Cox)
- Study CDS pricing and curve bootstrapping
- Understand default correlation and copula models
- Learn jump-to-default modeling
- Study CDO tranche pricing

4.4 Success Metrics - Phase 4

You're ready for Phase 5 if you can:

- Calibrate Hull-White model to market cap/floor prices
- Explain why SABR is better than Black-Scholes for swaptions
- Price an FX option using Garman-Kohlhagen and adjust for quanto if needed
- Implement Heston model and explain vol-of-vol impact
- Bootstrap a CDS curve and price a first-to-default basket
- Articulate when each model is appropriate and what risks it ignores
- Explain model risk for each asset class with concrete examples

5 Phase 5: Derivatives Products (Weeks 21–28)

Objective: Learn the complete product lifecycle from payoff definition to model selection to P&L attribution for 57 derivative products across all asset classes.

Critical Note: Study products AFTER models for each asset class. This ensures you understand WHY certain models are chosen for specific products.

5.4 Success Metrics - Phase 5

You're ready for Phase 6 if you can:

- Explain the payoff structure of any of the 57 products to a non-quant
- Choose the appropriate pricing model for each product and justify your choice
- Calculate Greeks and design a hedging strategy for vanilla and exotic products
- Explain P&L attribution for a portfolio containing multiple products
- Identify model risk for each product and suggest model improvements
- Describe how market conditions affect product pricing and hedging
- Answer product-related interview questions with confidence

6 Phase 6: Practical Implementation (Weeks 29–36)

Objective: Build 43 industry-grade projects that demonstrate your ability to implement pricing models, risk systems, and trading strategies.

Why Projects Matter: Interviews increasingly include take-home assignments and coding assessments. A strong GitHub portfolio with production-quality code differentiates you from other candidates.

6.1 Resource

1. Quant Projects Pack (43 Projects) (PROJECT10)

Industry-grade projects across three difficulty levels: 10 Foundational, 10 Moderate (Core Pricing Methods), 10 Advanced, plus 13 asset-class-specific implementations. Includes full code, documentation, testing frameworks, and performance optimization techniques.

→ [Click Here to Access](#) — Coupon: PROJECT10

6.2 Project Structure

Foundational Level (10 projects):

- Project 1: One-Period Binomial Model and Arbitrage Detection
- Project 2: GBM Simulation and Monte-Carlo Pricing of European Options
- Project 3: Historical and EWMA Volatility Estimation
- Project 4: American Put Pricing via Binomial Model
- Project 5: Barrier Hitting Probability Simulation
- Project 6: Asian Option Pricing via Monte-Carlo
- Project 7: Euler-Maruyama Simulation of Stochastic Differential Equations
- Project 8: Implied Volatility Computation Using Newton's Method
- Project 9: Exchange Option Pricing via Monte-Carlo Simulation
- Project 10: Numerical Greeks via Finite Differences

Moderate Level - Core Pricing Methods (10 projects):

- Project 1: Multi-Step Binomial Tree for European, American, and Exotic Options
- Project 2: Trinomial Tree with Adaptive Branching and Barrier Support
- Project 3: Crank-Nicolson Finite-Difference PDE Solver with Boundary Conditions
- Project 4: Robust Finite-Difference Greeks with Richardson Extrapolation
- Project 5: Merton Jump-Diffusion Monte-Carlo with Variance Reduction
- Project 6: Heston Stochastic Volatility Monte-Carlo with Exact Simulation
- Project 7: Local Volatility Surface via Dupire Formula
- Project 8: Implied Volatility Surface Construction and Arbitrage-Free Smoothing
- Project 9: American Barrier Option Pricing via Monte-Carlo Regression
- Project 10: Calibration of Heston Model to Market IV Surface

Advanced Level (10 projects):

- Project 1: Hull-White One-Factor Model Calibration

- Each project should take 6-8 hours

Week 36: Advanced Projects (10 projects)

- Complete 1-2 advanced projects per day
- Focus on optimization, performance, real-time capabilities
- Integrate multiple components into unified frameworks
- Each project should take 8-12 hours

6.4 Best Practices for Project Implementation

1. **Version Control:** Use Git from day one, commit frequently with clear messages
2. **Code Structure:** Separate data, models, analytics, and visualization layers
3. **Documentation:** Write clear README files with usage examples and mathematical background
4. **Testing:** Include unit tests comparing your results to analytical solutions
5. **Performance:** Profile your code, optimize bottlenecks, use vectorization
6. **Visualization:** Create clear plots for Greeks, P&L, convergence analysis
7. **Error Handling:** Implement robust error checking and edge case handling
8. **Extensibility:** Design code that can be easily extended for variations

6.5 Programming Languages & Tools

Primary Languages:

- **Python:** NumPy, SciPy, Pandas, Matplotlib for rapid prototyping
- **C++:** For performance-critical components (advanced projects)
- **R:** For statistical analysis and econometrics projects

Essential Libraries:

- QuantLib for derivatives pricing
- TensorFlow/PyTorch for ML projects
- Jupyter notebooks for interactive analysis
- Git/GitHub for version control and portfolio

6.6 Success Metrics - Phase 6

You're ready for Phase 7 if you:

- Have completed 40+ projects with clean, documented code
- Maintain a GitHub portfolio that showcases your work
- Can explain design decisions and trade-offs in your implementations
- Have tested your code against analytical solutions and benchmark results
- Can optimize code for performance (runtime, memory usage)
- Understand when to use Python vs C++ for different problems
- Can walk through your code in a technical interview setting
- Have integrated multiple components (e.g., pricing + risk + P&L)

7 Phase 7: Interview Mastery (Weeks 37–40)

Objective: Master 250+ interview questions covering mathematics, products, models, coding, and behavioral topics. Prepare for technical screens, take-home assignments, and on-site interviews.

7.1 Resource

1. 250+ Quant Interview Problems (FIRST10)

Comprehensive collection with solved examples and practice sheets. Covers probability, stochastic calculus, derivatives pricing, brainteasers, coding challenges, and behavioral questions. Organized by difficulty and topic.

→ [Click Here to Access](#) — Coupon: FIRST10